

Math 133 Written Homework 4 Solutions

Week 6: Limits and Continuity

1. (a) Polynomials are continuous everywhere, and the denominator at -2 is $(-2)^3 + 10 = 2 \neq 0$. The quotient law therefore gives

$$\lim_{x \rightarrow -2} \frac{3x^2 + x - 4}{x^3 + 10} = \frac{12 - 2 - 4}{-8 + 10} = \boxed{3}.$$

- (b) The linear factor and the square-root factor are continuous at 5 (the radicand is $9 > 0$). Both limits are finite, so the product law gives

$$\lim_{x \rightarrow 5} (2x - 3)\sqrt{x + 4} = (10 - 3)\sqrt{9} = \boxed{21}.$$

2. Both substitutions give the indeterminate form $0/0$.

- (a) For $x \neq 4$, $(x^2 - 16)/(x - 4) = x + 4$. Since this polynomial is continuous at 4 , the limit is $\boxed{8}$.

- (b) For $x \neq -2, 2$,

$$\frac{x^2 + 5x + 6}{x^2 - 4} = \frac{(x + 2)(x + 3)}{(x + 2)(x - 2)} = \frac{x + 3}{x - 2}.$$

The simplified quotient is continuous at -2 , since its denominator is $-4 \neq 0$. The limit is $\boxed{-\frac{1}{4}}$.

3. Both substitutions give the indeterminate form $0/0$.

- (a) For $x \neq 7$ near 7 , rationalizing gives

$$\frac{\sqrt{x + 2} - 3}{x - 7} = \frac{x + 2 - 9}{(x - 7)(\sqrt{x + 2} + 3)} = \frac{1}{\sqrt{x + 2} + 3}.$$

The last expression is continuous at 7 : its radicand is positive and its denominator is $6 \neq 0$. The limit is $\boxed{\frac{1}{6}}$.

- (b) For $x \neq 2$ near 2 ,

$$\frac{\frac{1}{x+3} - \frac{1}{5}}{x - 2} = \frac{2 - x}{5(x + 3)(x - 2)} = -\frac{1}{5(x + 3)}.$$

The simplified denominator is $25 \neq 0$ at 2 , so continuity of this rational function gives the limit $\boxed{-\frac{1}{25}}$.

4. (a) As $x \rightarrow 2^+$, both terms tend to $+\infty$, giving the indeterminate form $\infty - \infty$. For $x > 2$,

$$\frac{1}{x-2} - \frac{4}{x^2-4} = \frac{x+2-4}{(x-2)(x+2)} = \frac{1}{x+2}.$$

The simplified function is continuous at 2, with nonzero denominator 4, so the limit is $\boxed{\frac{1}{4}}$.

- (b) As $x \rightarrow 3^+$, the factors tend to 0 and $+\infty$, giving the indeterminate form $0 \cdot \infty$. For $x > 3$,

$$(x-3)\frac{x+2}{x^2-9} = \frac{x+2}{x+3}.$$

The simplified rational function has denominator $6 \neq 0$ at 3, so its continuity gives the limit $\boxed{\frac{5}{6}}$.

5. For $x \neq 1$, combine the terms before taking the limit:

$$\frac{1}{x-1} + \left(\frac{x+1}{x-1}\right)^2 = \frac{x-1 + (x+1)^2}{(x-1)^2} = \frac{x^2 + 3x}{(x-1)^2}.$$

The numerator tends to 4 by polynomial continuity, while the denominator tends to 0 through positive values. Thus the quotient tends to $+\infty$ from both sides, and

$$\lim_{x \rightarrow 1} \left[\frac{1}{x-1} + \left(\frac{x+1}{x-1}\right)^2 \right] = +\infty.$$

From the left, the original terms have the indeterminate form $-\infty + \infty$; from the right, both terms tend to $+\infty$. The combined expression resolves the left-hand case. The finite-limit quotient law does not apply when the denominator limit is zero.

6. The square root requires $x \geq -4$ and the denominator requires $x \neq 1$, so

$$\text{dom}(f) = [-4, 1) \cup (1, \infty).$$

The square-root numerator is continuous on $[-4, \infty)$ and the denominator is continuous and nonzero on this domain. Hence f is continuous throughout its domain, with right-hand continuity at -4 . In particular, 0 is in the domain, so

$$\lim_{x \rightarrow 0} f(x) = f(0) = \frac{\sqrt{4}}{-1} = \boxed{-2}.$$

For g , the logarithm requires $x - 2 > 0$. This already ensures $x + 1 \neq 0$, so

$$\text{dom}(g) = (2, \infty).$$

The logarithm is continuous for positive arguments, and the linear denominator is nonzero for $x > 2$. Thus g is continuous throughout $(2, \infty)$, and

$$\lim_{x \rightarrow 3} g(x) = g(3) = \frac{\ln 1}{4} = \boxed{0}.$$

7. The domain is $\mathbb{R} \setminus \{-2\}$. Since

$$h(x) = \begin{cases} -1, & x < -2, \\ 1, & x > -2, \end{cases} \quad \lim_{x \rightarrow -2^-} h(x) = -1, \quad \lim_{x \rightarrow -2^+} h(x) = 1,$$

h has a jump discontinuity at -2 . The one-sided limits follow from the constant values on either side. The first continuity condition fails because $h(-2)$ is undefined; the second fails because the two-sided limit does not exist. Consequently the third condition, equality of the limit and the function value, cannot hold. The function is continuous at every point of its domain.

8. Both functions have domain $\mathbb{R}$.

For A , the function value is $A(3) = 5$. Polynomial continuity of the two branches gives

$$\lim_{x \rightarrow 3^-} A(x) = 3^2 - 2 = 7, \quad \lim_{x \rightarrow 3^+} A(x) = 2(3) + 1 = 7.$$

Thus $\lim_{x \rightarrow 3} A(x) = 7 \neq A(3)$. The first two continuity conditions hold, but the third fails: the limit is not the function value. This is a removable discontinuity; replacing $A(3)$ by 7 makes A continuous at 3 .

For B , the function value is $B(3) = 2(3) - 1 = 5$. Continuity of the linear branches gives

$$\lim_{x \rightarrow 3^-} B(x) = 5, \quad \lim_{x \rightarrow 3^+} B(x) = 3 + 4 = 7.$$

The one-sided limits disagree, so the two-sided limit does not exist. This is a jump discontinuity. The first continuity condition holds, but the second fails; consequently the third cannot hold either. Changing only $B(3)$ cannot remove the jump.

9. The domain of p is $\mathbb{R}$. As a polynomial, p is continuous everywhere, in particular on $[1, 2]$. Also,

$$p(1) = 1 + 2 - 6 = -3, \quad p(2) = 8 + 4 - 6 = 6.$$

Since 0 lies strictly between $p(1)$ and $p(2)$, the Intermediate Value Theorem guarantees a $c \in (1, 2)$ such that $\boxed{p(c) = 0}$.

10. For every real k , the domain is $\mathbb{R}$ and each branch is a polynomial, so the only possible discontinuity is at 2. Polynomial continuity gives

$$H_k(2) = 4k + 1, \quad \lim_{x \rightarrow 2^-} H_k(x) = 4k + 1, \quad \lim_{x \rightarrow 2^+} H_k(x) = 7.$$

Continuity at 2 requires $4k + 1 = 7$, hence $\boxed{k = \frac{3}{2}}$.

For this value, $H_{3/2}(2) = 7$ is defined; both one-sided limits equal 7, so the two-sided limit exists; and that limit equals $H_{3/2}(2)$. All three continuity conditions hold. At every $x < 2$ or $x > 2$, the function agrees locally with a polynomial and is therefore continuous. Thus $k = \frac{3}{2}$ is the unique value giving continuity on all of $\mathbb{R}$.